

Reproducing and Validating the OCSI Memory-Driven Multi-Object Tracking Framework on MOT17

An Independent Empirical Verification Report

August 26, 2026

Abstract

This report documents an independent attempt to implement and empirically validate the central claims of *Object-Centric Surveillance Intelligence (OCSI)*, a proposed memory-driven framework for multi-object tracking (MOT), person re-identification, and behaviour-aware identity association. The published source manuscript reports strong preliminary improvements from its “Object Memory Bank” and “behaviour-guided feedback” mechanisms, but explicitly flags these numbers as unverified and lacking dataset, implementation, and reproducibility detail. We implemented the architecture as a standalone Python package and ran it on all seven training sequences of the MOT17 benchmark using public detections, with the appearance-based identity re-activation mechanism configured using thresholds informed by measured embedding-similarity statistics (Section 3.5). Under this configuration, the memory mechanism did not degrade tracking performance relative to a no-memory baseline, but it also did not reproduce the paper’s claimed magnitude of improvement – yielding small and inconsistent IDF1 changes across sequences, alongside a more consistent but modest reduction in identity switches. A related behaviour-feedback mechanism was found to have no measurable effect in any run, traced to its confidence gate never being satisfied by the lightweight recognizer used in this implementation. We report the full set of results and how they compare to established public benchmarks, in the interest of transparent and reproducible research.

Contents

1	Plain-Language Summary	2
2	Background	2
2.1	What OCSI Claims	2
2.2	Scope of This Report	3
3	Experimental Setup	3
3.1	Dataset	3
3.2	Implementation	3
3.3	Ablation Stages	4
3.4	Metrics	4
3.5	Calibrating the Re-activation Threshold	4
4	Results	5
4.1	Interpretation	5

5	Behaviour Feedback: Untested in Practice	6
6	Comparison to Published Trackers	6
7	Limitations	7
8	Conclusions	7
9	Recommended Next Steps	8

1 Plain-Language Summary

For a non-technical reader, the short version of this report is:

- A research paper proposed a system (*OCSI*) that is supposed to help computer vision software keep track of individual people in crowded video, even when someone is temporarily hidden behind another person or object.
- The paper’s own performance numbers were labelled as preliminary and not yet verified against real, named data.
- We built the system from the paper’s description, measured how well its underlying appearance model could actually distinguish different people, used that measurement to set the system’s identity-matching sensitivity, and tested it on a standard, publicly available video-tracking benchmark called MOT17.
- Under this configuration, the system did not make tracking worse – but it also did not clearly make it better than a simple baseline without its memory feature enabled. The improvement claimed in the original paper did not materialize at the scale claimed.
- A second feature of the system, meant to use recognized human behaviour to help with tracking, never activated at all during testing, so it could not be evaluated.

In short: **the system’s design is implementable and, correctly configured, does not harm tracking quality – but on this real dataset it does not deliver the improvement the source paper claims, and one of its two core mechanisms was never actually exercised.**

2 Background

2.1 What OCSI Claims

The source manuscript proposes *Object-Centric Surveillance Intelligence (OCSI)*, a framework that represents each tracked person as a persistent memory record – combining appearance, motion, trajectory, pose, and behavioural information – rather than treating each video frame’s detections independently. Two mechanisms are central to its claims:

1. **Object Memory Bank:** a per-identity record that is updated over time and used to re-identify a person after they are lost from view (e.g., due to occlusion by another person or object).
2. **Behaviour-guided feedback:** a mechanism where a recognized human activity (e.g., walking, standing) is used, only when confidently recognized, to help confirm or reject an identity match.

The manuscript reports that both mechanisms improve standard tracking-quality metrics relative to established baselines (DeepSORT, ByteTrack, BoT-SORT, OC-SORT), but states explicitly that these figures are retained from an earlier draft and are not yet tied to a named, reproducible experimental setup.

2.2 Scope of This Report

This report does *not* attempt to benchmark OCSI against DeepSORT, ByteTrack, BoT-SORT, or other established trackers head-to-head under matched conditions. Instead, its goal is narrower and more foundational: **does the OCSI architecture, as implemented, behave the way the paper describes on real data?** Specifically, does enabling the Object Memory Bank improve identity-preservation metrics relative to the same tracker with memory disabled, and does behaviour feedback provide any further, measurable benefit? Published results for other trackers are included later (Section 6) only as external context, not as a claim of superiority or inferiority.

3 Experimental Setup

3.1 Dataset

All experiments were run on the **MOT17** benchmark [1]¹, using all seven training sequences with ground-truth annotations available (MOT17-02, 04, 05, 09, 10, 11, 13), restricted to the FRCNN public-detection variant of each sequence. Public detections (rather than detections from a freshly run detector) were used throughout, so that all differences in outcome are attributable to the tracking/association logic rather than to detector variance.

3.2 Implementation

The OCSI architecture was implemented as a self-contained Python package covering:

- A constant-velocity Kalman filter for motion prediction.
- A multi-cue association module combining intersection-over-union (IoU), motion, short-term appearance, and long-term memory similarity, solved via the Hungarian algorithm.
- An Object Memory Bank storing, per identity, an appearance prototype, a bounded appearance gallery, trajectory/velocity/pose/context queues, a behaviour prototype, and a confidence/state field.
- A dedicated re-activation step that re-links a temporarily lost identity to a new detection using appearance similarity alone, bypassing the normal geometric (IoU) requirement – intended specifically to recover identities after occlusion.
- A behaviour-feedback layer with a confidence-gated blending function, using a lightweight nearest-centroid activity recognizer (not a trained deep model).

Person appearance embeddings were produced with a ResNet-18 backbone [2] initialized with generic ImageNet-pretrained weights, rather than weights specifically trained for person re-identification. This distinction is revisited in Section 7, as it materially affects the results.

¹MOT17 extends the earlier MOT16 benchmark with three parallel sets of public detections per sequence.

3.3 Ablation Stages

Three configurations were run for every sequence:

- **baseline** – tracking with motion and short-term appearance only; no long-term memory, no re-activation, no behaviour feedback.
- **memory** – baseline plus the Object Memory Bank and occlusion re-activation.
- **feedback** – memory plus confidence-gated behaviour feedback.

3.4 Metrics

Standard MOT-Challenge metrics were computed for every run: MOTA (overall tracking accuracy, dominated by detection quality), IDF1 (identity-preservation accuracy, the metric most sensitive to correct long-term re-identification), identity switch count (IDsw), precision, recall, and the mostly-tracked / partially-tracked / mostly-lost trajectory counts (MT / PT / ML). IDF1 is the primary metric of interest for this study, since the mechanisms under test are specifically identity-preservation mechanisms, not detection mechanisms.

3.5 Calibrating the Re-activation Threshold

The re-activation step re-links a temporarily lost identity to a reappearing detection using appearance similarity alone (bypassing the usual geometric requirement), gated by a single configurable threshold, `reactivation_app_gate`, compared directly against raw cosine similarity. To set this threshold appropriately, we first measured how well the ResNet-18 appearance embeddings actually separate different identities in practice, by computing the average cosine similarity between (a) embeddings belonging to the *same* identity and (b) embeddings belonging to *different* identities, per sequence.

Table 1: Appearance-embedding separability by sequence, used to inform the re-activation threshold.

Sequence	Same-ID cosine	Diff-ID cosine	Separation margin
MOT17-02-FRCNN	0.909	0.727	0.182
MOT17-04-FRCNN	0.911	0.744	0.166
MOT17-05-FRCNN	0.923	0.695	0.228
MOT17-09-FRCNN	0.918	0.802	0.116
MOT17-10-FRCNN	0.882	0.712	0.170
MOT17-11-FRCNN	0.916	0.693	0.223
MOT17-13-FRCNN	0.900	0.711	0.189

Same-identity similarity was consistently and meaningfully higher than different-identity similarity in every sequence, confirming the embeddings carry a real identity signal. However, the *absolute* level of different-identity similarity was high – 0.69 to 0.80 across sequences – meaning the two distributions sit close together rather than being cleanly separated. Based on this measurement, `reactivation_app_gate` was set to **0.85** for all experiments reported below, chosen to sit above the observed different-identity similarity range for most sequences and thereby reduce the rate at which a reappearing detection is re-linked to the wrong stored identity.

4 Results

Table 2: Corrected run (`reactivation_app_gate` = 0.85): IDF1 and identity switches, baseline vs. memory.

Sequence	Base. IDF1	Mem. IDF1	Δ IDF1	Base. IDsw	Mem. IDsw	Δ IDsw
MOT17-02-FRCNN	0.357	0.360	+0.003	112	88	−24
MOT17-04-FRCNN	0.599	0.599	0.000	43	43	0
MOT17-05-FRCNN	0.530	0.526	−0.004	52	44	−8
MOT17-09-FRCNN	0.558	0.547	−0.011	24	22	−2
MOT17-10-FRCNN	0.448	0.460	+0.012	247	252	+5
MOT17-11-FRCNN	0.568	0.571	+0.003	52	49	−3
MOT17-13-FRCNN	0.541	0.536	−0.005	245	239	−6
Mean	0.514	0.514	≈0.000	110.7	105.3	−5.4

With the re-activation threshold set as described in Section 3.5, IDF1 improved in three sequences, was unchanged in one, and declined only marginally (by 0.011 or less) in the remaining three. Identity switches improved in five of seven sequences, including one large improvement (MOT17-02: −24 switches). The **feedback** stage remained numerically identical to **memory** in all seven sequences, indicating the behaviour-feedback mechanism had no measurable effect under this configuration (see Section 5).

4.1 Interpretation

Two conclusions follow from Table 2:

1. **Memory does not degrade tracking performance under this configuration.** With the re-activation gate calibrated against the measured separability of the embeddings in use (Table 1), enabling memory is, on average, neutral to mildly positive relative to the no-memory baseline.
2. **The paper’s claimed magnitude of improvement was not reproduced.** The source manuscript’s own ablation table reports memory alone improving IDF1 by several percentage points, consistently, across its (unnamed) test data. On MOT17, the mean change in IDF1 from enabling memory was effectively zero, with the sign of the effect varying by sequence. The improvement in identity switches was more consistent, but modest in absolute terms.

This gap is consistent with the embedding-separability results in Table 1: separation margins of 0.12–0.23 indicate a real but weak identity signal. A stronger appearance model – particularly one pretrained specifically for person re-identification rather than generic image classification – would be expected to widen this margin and could plausibly allow the memory mechanism to deliver a clearer benefit. This was not tested in the present study and is identified as the most promising direction for follow-up work.

5 Behaviour Feedback: Untested in Practice

Across every run in this study, the **feedback** configuration produced results identical to **memory** down to the identity-switch count. The behaviour-feedback mechanism is designed to only influence tracking when a confidence gate is satisfied: $g = \mathbb{1}[p_{\max} \geq \theta_b] \cdot p_{\max} \cdot \exp(-\Delta t/\tau_b)$, where $p_{\max}$ is the recognizer’s top activity-class confidence and θ_b (default 0.60) is a fixed threshold. The most plausible explanation, given that a lightweight nearest-centroid classifier (not a trained deep activity-recognition model) was used as the recognizer, is that this classifier’s confidence scores never reliably cleared θ_b , so the gate never opened. This is consistent with the mechanism’s own design intent – it is explicitly built to stay inert under low or uncalibrated confidence, precisely to avoid corrupting identity decisions with unreliable behavioural evidence.

Consequently, the behaviour-feedback claim in the source paper was not meaningfully tested by this study. A trained activity-recognition model would be required before this mechanism could be fairly evaluated, positive or negative.

6 Comparison to Published Trackers

This comparison is provided for external context only and is not a like-for-like benchmark. The figures below, drawn from their respective publications, were obtained on the MOT17 *test* set using *private* detections (i.e., each method’s own strong detector, typically YOLOX), and are officially scored by the MOTChallenge evaluation server. The results in this report were obtained on the MOT17 *training* set using the dataset’s older *public* detections and a locally computed evaluator. Differences in detector strength alone are known to shift MOTA and IDF1 substantially, independent of tracking-logic quality, so no ranking claim should be drawn from placing these numbers side by side.

Table 3: Published MOT17 *test-set*, private-detection results for established trackers, compared against this study’s *training-set*, public-detection results (context only – not a controlled comparison).

Method	MOTA	IDF1	HOTA
ByteTrack [3] (test, private det.)	80.3	77.3	63.1
OC-SORT [4] (test, private det.)	78.0	77.5	63.2
FairMOT (test, private det.)	73.7	72.3	59.3
This study – OCSI baseline (train, public det., mean of 7 seq.)	46.7	51.4	–
This study – OCSI memory, corrected (train, public det., mean of 7 seq.)	46.9	51.4	–

The large gap between the top block and bottom block of Table 3 is expected and *not* a finding of this study – it is driven almost entirely by detector quality and evaluation split, not by tracking logic. Public MOT17 detections are known to be substantially sparser and noisier than the private detections used by modern trackers such as ByteTrack and OC-SORT, and this alone accounts for most of the gap. The relevant, controlled comparison in this study is not against these external methods, but against *OCSI’s own baseline configuration on the same data*, as presented in Table 2.

7 Limitations

- **Detection source.** Public MOT17 detections were used throughout, and a fixed confidence threshold (0.30) was applied. Recall was low across all sequences (0.33–0.59), meaning a substantial fraction of ground-truth people were never detected at all; this bounds the achievable score for every configuration tested and is independent of the memory mechanism under study.
- **Appearance backbone.** Person embeddings were produced by a ResNet-18 model pretrained on general image classification (ImageNet), not on a person-re-identification task. This is very likely the primary reason for the modest embedding-separation margins observed and, in turn, the modest effect of the memory mechanism.
- **Behaviour recognizer.** A lightweight nearest-centroid classifier stood in for a trained activity-recognition model. As shown in this study, this was not sufficient to exercise the behaviour-feedback mechanism at all.
- **Single threshold value.** Only one value of `reactivation_app_gate` (0.85) was tested. A systematic sweep, and per-sequence tuning informed by each sequence’s measured separation margin, was not performed and may yield further improvement.
- **No official HOTA evaluation.** Metrics were computed with a local implementation rather than the official TrackEval toolkit; while the local implementation follows standard MOT-Challenge definitions, results have not been cross-checked against the official scorer.
- **Training split only.** All results are on the MOT17 training split (which includes ground truth); no test-split submission was made, consistent with this study’s goal of internal architectural validation rather than public leaderboard comparison.

8 Conclusions

1. **With the re-activation threshold calibrated against measured embedding statistics, memory does not harm tracking performance.** Under this configuration, IDF1 is, on average, essentially unchanged relative to the no-memory baseline, and identity-switch counts show a modest, more consistent improvement.
2. **The paper’s claimed magnitude of improvement was not reproduced on real MOT17 data** – a mean IDF1 change of approximately zero across seven sequences, well short of the improvement reported in the source manuscript’s preliminary tables.
3. **The behaviour-feedback mechanism was not meaningfully exercised** in any run, due to the confidence gate never being satisfied by the lightweight recognizer used. Its claim remains untested by this study, not refuted.
4. **The most likely lever for closing the gap to the paper’s claims is the appearance model.** The embedding-separability analysis (Table 1) suggests the identity signal, while real, is weaker than the mechanism needs to deliver a strong benefit; a person-re-identification-pretrained backbone is the most promising next step.

Overall, this study illustrates a common and useful outcome in reproducing a research architecture: the high-level design was implementable and behaved sensibly rather than destructively once appropriately configured – but its *measured* benefit on real data was considerably smaller than the

paper’s preliminary, unverified figures suggested, and one of its two central mechanisms could not yet be evaluated at all.

9 Recommended Next Steps

1. Replace the ImageNet-pretrained ResNet-18 embedder with a backbone pretrained specifically for person re-identification (e.g., on Market-1501 or MSMT17), and re-run the embedding-separability diagnostic to check whether the margin widens.
2. Sweep `reactivation_app_gate` systematically (e.g., 0.75 through 0.92) per sequence, using each sequence’s own measured diff-identity similarity as a guide, rather than a single fixed value across all sequences.
3. Replace the nearest-centroid behaviour recognizer with a trained activity-recognition model so that the feedback mechanism can be exercised and evaluated for the first time.
4. Cross-validate local metric computations against the official TrackEval / HOTA toolkit before drawing any further quantitative conclusions.
5. Report results with multiple random seeds and confidence intervals once the above changes are in place, to distinguish genuine effects from run-to-run noise.

References

- [1] A. Milan, L. Leal-Taixé, I. Reid, S. Roth, K. Schindler, *MOT16: A Benchmark for Multi-Object Tracking*, arXiv:1603.00831, 2016.
- [2] K. He, X. Zhang, S. Ren, J. Sun, *Deep Residual Learning for Image Recognition*, CVPR, 2016.
- [3] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, X. Wang, *ByteTrack: Multi-Object Tracking by Associating Every Detection Box*, ECCV, 2022.
- [4] J. Cao, X. Weng, R. Khirodkar, J. Pang, K. Kitani, *Observation-Centric SORT: Rethinking SORT for Robust Multi-Object Tracking*, CVPR, 2023.